

Rdrop main-board functional review

Reviewed 23 September 2026. **Disposition: do not release the present design for fabrication as a verified 15 A continuous board.** Several circuit corrections are needed independently of copper width. This is a design review, not a tested prototype qualification.

The adjacent `PCB Design - Review` project contains review annotations only. Its electrical connections and copper are unchanged. The annotations identify required changes; they are not evidence that those changes have been implemented. Original source, manufacturing and mechanical files are preserved. See `verification.json` for the preservation and copy checks.

Scope and evidence

Reviewed the current eight-sheet schematic, exported connectivity, PCB pads/tracks/zones, BOM, reference drawings, and the available mechanical STEP file headers/assembly context. Ran KiCad schematic ERC and PCB DRC, including schematic parity. The evidence folder contains the netlist, check reports and extracted PCB geometry. The existing STEP files are accessible locally, including `Mechanical/ard.mb-02.01.00.step` and the `_withPM.step` version; neither has been changed. A native CAD interference/clearance analysis has **not** been performed.

Design assumptions supplied by the owner: two layers, 1 oz copper, JLCPCB manufacture; 15 A continuous capacity with at most 15–20°C copper temperature rise; 18S / 78 V input; custom APM rated 5 V / 20 A continuously; Tiankong RC H20/35 servo with approximately 7 A peaks at 7.4 V; Hobbywing UBEC-10A-HV(3-14S)-V1. The PCB file specifies

1.6 mm thickness. APM continuous rating is accepted as owner-supplied, not independently certified here.

Missing evidence: Butiks (1) .DXF, full casing/gear assembly and tolerances, exact servo manufacturer specification, APM schematic/protection/efficiency and test record, firmware, exact ratings for several sourced parts. These prevent a complete mechanical, firmware and system-level sign-off. Findings below distinguish confirmed connectivity faults from parts requiring qualification.

What the power circuit actually does

Path	Connectivity in the current source
Drone battery	CN1 pins 4–9 → BATT+ → U302 conductive standoff → APM input; CN1 pins 13–18 and U303 are ground
APM output	U304 supplies +5V_1; U305 is its ground. +5V_1 → F2 → SRV+; separately +5V_1 → D302 → +5V logic rail
UBEC output	J1 pin 2 → +UBEC_OUT → F1 → SRV+; separately +UBEC_OUT → D301 → U301 LM1084-5.0 → D303 → +5V logic rail
Logic	+5V → U306 BL8072-3.3 → ESP32/CAN/IR/encoder. CH343 and LM358 also take power from +5V
Servo return	J2 pin 1 → SRV– → 0.501, a 0.1 Ω shunt → board ground

The LM1084 does not carry servo current in this schematic. It regulates the UBEC-fed logic branch. The APM bypasses it as intended. The logic net called +5V is lower than 5 V by D302 or D303’s forward drop. U302–U305 are conductive mechanical standoffs, not semiconductor devices.

Corrections required for functionality

1. Replace the specified UBEC for an 18S / 78 V installation

Confirmed mismatch. Hobbywing specifies the exact 10A HV model for 3–14S and 10 A continuous; 25 A is instantaneous. It therefore fails both the stated input requirement and the 15 A continuous output requirement. Select a converter qualified for the actual maximum battery voltage plus measured transients, 7.4 V output, and at least the required continuous load at enclosure temperature. A nominal 80 V limit leaves only 2 V against the stated 78 V maximum; it needs explicit transient assessment. A suitably designed 100 V-class input stage is a more useful starting specification, not automatic proof of suitability. [Hobbywing specification](#)

The APM's owner-confirmed 20 A continuous rating makes the **5 V route the more direct starting point**, provided the servo's permitted voltage and delivered torque/speed at 5 V are established. A module's output rating does not qualify the PCB, standoffs, fuse or servo connector.

2. Make the 5 V / 7.4 V selection mutually exclusive

Confirmed connectivity hazard if both source paths are enabled. F1 and F2 join at SRV+. Fuses conduct in both directions; they do not isolate supplies. With both fitted and both sources present, 7.4 V can feed the APM's 5 V output through F2 and then the logic rail through D302. D302/D303 do not isolate this bypass route. An unpowered converter can also be backfed.

For a fixed assembly variant, populate only the selected servo path, explicitly mark the other DNP in the schematic, PCB and assembly BOM, and label the permitted configuration. For field selection, use a break-before-make selector or reverse-current-blocking power switch arrangement rated for the full continuous current, inrush and voltage. Avoid ordinary small signal jumpers. If both converters may remain connected for logic power, confirm each permits its resulting unpowered/output conditions.

The existing fabrication BOM lists **F2 only**, but the PCB does not mark

F1 DNP. This may represent an intended APM-only build; resolve the inconsistency rather than assuming both fuses are physically assembled. Updating a BOM alone is inadequate protection against an incorrectly configured assembly.

3. Replace the servo shunt and redesign the current measurement

Confirmed electrical failure at the target current. Reference 0.501, RA2512F3R100G, is 0.1 Ω. Its listed 3 W power rating still requires primary manufacturer confirmation, but the calculated dissipation already demonstrates the problem:

Servo current	Shunt voltage loss	Shunt heat
7 A	0.70 V	4.90 W
15 A	1.50 V	22.50 W

With a nominal 3 W rating, $\sqrt{(3/0.1)} = 5.48$ A is the theoretical underated DC limit, not a safe design current. At 7 A the nominal 7.4 V servo supply becomes approximately 6.7 V before other wiring losses. The raised servo ground also reduces a 3.3 V PWM high to approximately 2.6 V relative to the servo at 7 A, or 1.8 V at 15 A.

R501/R502 attenuate the shunt signal by 10/11; R503/R504 set LM358 gain to 11. The net gain is 10, giving an ideal **1 V/A**. Thus 7 A demands 7 V output from an amplifier powered by less than 5 V, and the signal can exceed the ESP32 ADC range well before the requested full scale. Saturation is not overvoltage protection.

Recommended implementation: a **2 mΩ shunt, at least 3 W with its specified PCB thermal conditions**, four-terminal/Kelvin sensing, placed in the low-voltage servo positive rail, and **INA180A2 gain 50** supplied from 3.3 V. This yields 0.1 V/A: 0.7 V at 7 A, 1.5 V at 15 A and 2.0 V at 20 A. Shunt loss becomes 14 mV / 0.098 W at 7 A and 30 mV / 0.45 W at 15 A. Servo ground can then connect directly to common ground. Include 100 nF bypass, an ADC filter, input transient assessment and calibration. This amplifier belongs on the **5/7.4 V rail**,

never the 78 V input; its common-mode limit is 26 V. It is unidirectional: use a bidirectional alternative if regenerative current must be measured. Verify exact A/B package pinout. [TI INA180 datasheet](#)

The INA180's input offset makes very small currents relatively inaccurate with a 2 m Ω shunt; the proposed scaling prioritizes jam/overload measurement. Set software thresholds from calibrated current and time, not one instantaneous ADC sample. If retaining U501 for another purpose, terminate its unused B amplifier in a valid closed-loop configuration and add local decoupling; its second unit is presently absent from the schematic.

4. Qualify the entire 15 A path, including fuses and connectors

Unqualified and likely underspecified components. F1/F2 TLC-MSML750/12 have not been supported by a retrieved manufacturer datasheet. The apparent 7.5 A hold / 15 A trip designation must be verified; a 15 A trip rating is not a 15 A continuous rating. Obtain the hold-current derating, resistance, trip curves, voltage and interrupt/fault rating. Replace as necessary with protection coordinated to the cable, connector, copper and source. Do not simply install a larger fuse without upgrading the protected path. Any 12 V-rated protection belongs only on a low-voltage branch, not raw battery.

J1 and J2 use a single 2.54 mm power contact and a single return contact. No evidence establishes 15 A continuous capability for these exact parts. Replace the high-current connections with documented, suitably derated locking power connectors or strain-relieved soldered cable pads, including the servo pigtail. A 20 A-or-higher connector rating can provide margin only if its ambient, wire size and mating conditions match this installation.

Qualify the U304/U305 conductive standoffs for solder-joint/contact resistance, continuous current and mechanical fastening. Do not infer a current rating from their M3 size. At 15 A, **each 1 m Ω produces 0.225 W**. Contact resistance and local necks can dominate otherwise large pours.

CN1 is different: six BATT+ and six ground contacts are paralleled. JAE's published standard rating is 3 A/contact, with higher operation conditional on circumstances. At 15 A through this connector, ideal sharing would be 2.5 A/contact, requiring all six wired and sharing properly. In normal use it carries the **high-voltage converter input current**, not the 15 A low-voltage output current. Verify the exact mating housing, terminals, crimp/wire size, pin numbering viewed from each side, voltage rating and populated harness. [JAE MX23A manufacturer specification](#)

5. Connect AS5600 correctly and add I²C pull-ups

Confirmed fault. U801 pin 2 is on 3.3 V, while pin 1 VDD5V is marked no-connect. In 3.3 V operation **pins 1 and 2 must both connect to 3.3 V**. Remove the no-connect and route pin 1 accordingly. C801 = 100 nF is appropriate to retain locally for this mode. SDA/SCL lack external pull-ups in the exported netlist: add one per line to 3.3 V, initially 4.7 k Ω at 100 kHz, or select lower resistance from bus capacitance for faster operation. For example, 4.7 k Ω $\times$ 100 pF gives approximately 398 ns rise time using $0.8473RC$; 2.2 k Ω gives about 186 ns.

Use a diametrically magnetized magnet, correctly centered over the sensing location, and verify field/gap and magnet-status registers across mechanical tolerances. DIR grounded is valid. [AS5600 manufacturer datasheet](#)

6. Drive the IR emitters independently of the visible LEDs

Confirmed inadequate transmitter architecture. GPIO4 $\rightarrow$ D601 visible LED $\rightarrow$ D602 TSAL6200 $\rightarrow$ R603 1.2 k Ω $\rightarrow$ ground, and the analogous GPIO19/D604/D605/R606 branch, place both LEDs in series on 3.3 V. Current is approximately $\max(0, (V_{GPIO} - V_{Fvisible} - V_{FIR})/1200)$: little voltage remains, giving sub-milliampere or near-zero current depending on parts and temperature. Reliable detection across the opening is not established by this circuit.

Use one low-side MOSFET driver per TSAL6200, with a separate IR series resistor from the logic supply. Choose a MOSFET with

guaranteed conduction at 3.3 V gate drive, gate pull-down and modest series gate resistor. A **68 Ω , 0.5 W resistor from approximately 4.7–5 V** is an initial roughly 45–55 mA on-current design; verify LED forward voltage, supply tolerance and optical margin. Use separate indicators or omit them. About 50 mA on-current is a design choice, not a TSAL6200 consumption specification. Add local bypass/bulk capacitance and keep pulsed return current out of analog sensing. [TSAL6200 datasheet](#)

7. Remove the heavy load on the TSSP4038 output

Confirmed unsuitable loading. U601 output drives Q601 through R602 = 510 Ω and also a visible LED/R607 branch. The receiver's approximately 33 k Ω internal pull-up is not a power source for these loads. Its high level is substantially loaded; the LED is not a valid direct output load.

Connect the 3.3 V receiver output to GPIO35 directly, optionally with a suitably light external pull-up, removing Q601/R602 and the output LED load. Remove or repurpose R601 consistently. This changes the firmware polarity relative to the transistor-inverted design: receiver detection is active low. Send qualified 38 kHz bursts, at least ten carrier cycles, and evaluate missing/detected bursts with a timeout. Receiver delay is on the order of 7–13 carrier cycles. Verify the complete timing against the fastest bait passage; a constant DC LED does not exercise this receiver correctly. [TSSP4038 datasheet](#)

8. Make the analog IR node a sensor input rather than an indicator driver

Confirmed sensitivity/loading issue. Q602 has R604 = 220 Ω to 3.3 V; R605 = 220 Ω and D603 also load its collector. With an illustrative red LED drop of 1.9 V, the dark node is about 2.6 V and the indicator branch draws 3.18 mA. Strong illumination can demand $3.3/220 = 15$ mA from the pull-up. This wastes signal sensitivity and gives a nonlinear LED-loaded response.

Remove D603/R605 from the measurement node. Start R604 at 10 k Ω , then select approximately 4.7–47 k Ω from actual light/dark margin. With

10 k Ω , full pull-up current is only 0.33 mA. Use controlled emitter on/off sampling with ambient subtraction and time-separate the two optical channels to avoid cross-talk. For an initial slow on/off sampling scheme, C602 = 1 nF gives a 10 μ s nominal RC time constant; wait at least several time constants and verify phototransistor recovery. Do not expect the original 100 nF with a 10 k Ω resistor to resolve a 38 kHz carrier. ADC readings approaching 3.3 V may saturate the useful measurement range even though the pin voltage is within the rail; use calibration/scaling or threshold detection as appropriate. [TEFT4300 datasheet](#)

9. Fit only the CAN termination required by the installed bus

Confirmed duplicate termination. R701 and R702 are each 120 Ω and both populated on the same CANH/CANL pair: this board alone presents 60 Ω . With a normal far-end 120 Ω termination, the total becomes 40 Ω . For an endpoint, fit one 120 Ω resistor, preferably at the cable entry; for an intermediate node, fit neither. Mark the variants in the BOM. With the assembled bus unpowered, approximately 60 Ω across H/L is the expected result for two endpoint terminators in parallel. Add appropriate cable-interface protection and check ground offset/fault voltage; a 3.3 V CAN transceiver is not a 78 V fault-tolerant interface. [SN65HVD230 datasheet](#)

10. Establish regulator stability, headroom and enclosed thermal performance

Qualification required; not a claim that the servo overloads

LM1084. C302 is a 22 μ F 0603 MLCC. The HTC LM1084 stability guidance specifies 22 μ F solid tantalum or 150 μ F aluminum electrolytic for assured operation across conditions. An MLCC with the same printed capacitance is not automatically equivalent: effective capacitance and ESR differ. Use a manufacturer-supported capacitor network or a regulator explicitly stable with the selected ceramic network, then check startup and load steps. The available HTC document lists 1.5 V dropout at full 5 A; allow D301 loss and UBEC droop. [HTC LM1084 datasheet](#)

The BL8072 is a 2 A-rated LDO, not a guaranteed 2 A output in this enclosure. At a 4.7 V input and 0.5 A on 3.3 V it dissipates approximately 0.7 W. A UBEC-fed LM1084 at about 7.1 V input and 0.5 A adds approximately 1.05 W, before diode losses. If the claimed 2–3 A electronics load includes external equipment, identify and budget it explicitly; 3 A through BL8072 exceeds its nominal capability. A switching logic regulator becomes preferable if measured heat or external loading requires it. [BL8072 datasheet](#)

Verify exact BN5A2V30 forward loss, reverse rating, package heat dissipation and surge behavior from its manufacturer datasheet before release. It has not been retrieved reliably here. These three diodes carry logic-branch current, not the full servo current. Check MLCC voltage ratings and DC-bias capacitance on all regulator rails. Add bulk capacitance at the servo connection based on measured cable inductance and supply response: $C \geq I \cdot \Delta t / \Delta V$ is a first estimate, not a substitute for a regulator that supplies sustained load. Evaluate servo regeneration and rail overshoot; capacitor size/TVS selection needs measured energy and regulator sink behavior.

11. Isolate programming USB power and confirm control-signal levels

Confirmed shared USB/system power node. J3 pin 2, CH343 VDD and CH343 VBUS sense share +5V. An external programming adapter carrying host VBUS can therefore be fed by the powered board. Provide a suitable USB power-isolation arrangement and sense actual connector VBUS as required by the CH343 configuration, or define and enforce a data-only programming interface with correct VBUS sensing. Test powered/unpowered combinations. The CH343 has an internal clock source; a missing crystal is not a finding. [CH343 manufacturer datasheet](#)

Confirm CN1 ARM/FEED/COMMAND signal voltage, direction, ground reference and startup state against the drone controller. ESP32 GPIO is not a direct 5 V interface; add level conversion/protection if needed. R201 provides a 4.7 k Ω COMMAND pull-down; equivalent external pull-downs are not shown for every control input. Firmware/internal pulls alone may not define the state before boot. Define a hardware-safe servo startup state, disarm default, command timeout, jam limit and

recovery behavior. Check automatic programming/reset works under slow supply ramp; C203 = 100 nF with R202 = 10 k Ω gives only about 1 ms RC.

Q201/Q202 use custom CBE numbering with a correspondingly renumbered footprint. A standard SS8050 symbol/footprint substitution without checking the physical pad mapping can introduce a fault. The current mapping was inspected; the name alone is not evidence of a reversed transistor. Preserve or normalize symbol and footprint together.

12. Complete the raw-battery protection and insulation review

No dedicated raw-input fuse, reverse-polarity element or surge clamp is evident on this main board. Determine which protections already exist in the drone harness/APM, then add the missing functions at system level. Specify parts from maximum continuous voltage, fault current and measured switching/hot-plug transients. Do not apply a low-voltage servo fuse or the INA180 to BATT+. Audit spacing to grounded screws, standoffs, enclosure and adjacent signal copper for the actual 78 V environment, contamination and manufacturing tolerances. A low-voltage DRC clearance setting alone does not prove this.

Copper and mechanical assessment

Large pours are useful, but they do not establish the requested temperature rise through pads, thermal spokes, layer transitions, fuses, shunt and connectors. The saved DRC reports **ten starved-thermal errors**, on CN1 ground pins 14–18 on both copper layers: one usable spoke instead of the required two. Rework those local connections, refill and rerun DRC; verify solderability and current distribution. The narrow tracks visible on some supply nets include low-current branches, so their widths were not incorrectly treated as if all carried 15 A.

For a numerical screening comparison, the traditional external-trace relationship $I = 0.048 \cdot \Delta T^{0.44} \cdot A^{0.725}$, with A in mil², gives the following widths for nominal 35 μ m copper:

Current in one external copper layer	15°C rise screening width	20°C rise screening width
15 A	9.84 mm	8.26 mm
7.5 A, assuming ideal equal sharing of 15 A between two layers	3.78 mm per layer	3.18 mm per layer

This empirical trace calculation is **not** a temperature prediction for this shaped two-layer board. Equal sharing between layers requires suitable connections and similar resistance; two pours on the same net are not automatically equivalent to a single 2 oz conductor. Account for hot copper resistance, minimum manufactured thickness, nearby heat and the enclosed ambient. Use a physical 15 A thermal test for acceptance. [KiCad PCB calculator documentation](#)

The bait opening has approximately 42 mm between its straight sidewalls in the PCB geometry. The opposing digital pair's lead centers are approximately 71.0 mm apart; the analog pair's about 57.6 mm. These are placement measurements, not validated optical path lengths. Preserve the board edge and cutout until the actual casing and mechanism are registered together. The bottom-mounted encoder, servo connector and other parts need clearance to rotating gears and fasteners. The absent DXF prevents that check.

The TSAL6200 is a 5 mm leaded emitter. The custom footprint has a side-emitter name but modified geometry; check the physical right-angle lead forming, lens axis, height and mounting tolerance against the selected part. A footprint name alone does not prove incompatibility. Verify receiver/emitter polarity and optics from their package drawings, not 3D appearance alone.

Two crossing beams do not guarantee that every bait trajectory intersects a beam. Test the smallest bait, all orientations and off-center paths; add a guide or additional optical coverage if blind regions exist. Test sunlight, dust, vibration and partial obstruction. Confirm the magnet location and underside clearance using the actual assembled mechanism. None of these can be signed off from the isolated PCB STEP model.

Current consumption: what can and cannot be estimated

The following values distinguish datasheet operating current, calculated branch current and design allowances. Servo operating current depends on load, speed, holding torque and stall duration. **7 A at 7.4 V is the owner's supplied peak estimate**, not an independently verified Tiankong limit. Its 5 V stall/current/torque data remain unknown. A 15 A board capacity requirement does not mean the device normally consumes 15 A.

The current of a regulator, diode, fuse, shunt and connector carrying the same load must not be added repeatedly. Capacitors are not steady DC loads; charging/inrush and leakage are separate. With the existing linear logic regulators, the logic input current is approximately the sum of the load currents plus regulator ground current. With switching replacements, use power and efficiency instead.

Component/ reference	Current basis in current circuit	Corrected design / implication
U201 ESP32- WROOM-32E	RF TX example 239 mA average / 379 mA peak at 19.5 dBm; RX about 112–118 mA. Chip modem-sleep examples span about 20–68 mA, excluding module/board additions	Use 50 mA only as an RF-off planning example; reserve ≥500 mA capability for ESP32 supply transients. Same module remains
U202 CH343P	Manufacturer VDD operating table: 3 mA typical, 15 mA maximum; VIO supplies output loads separately, not another automatic 10 mA idle load	Keep 3 mA example; reserve 25 mA including possible VIO loading in a conservative board budget

U701 SN65HVD230	Approximately 10 mA typical / 17 mA maximum no-load operating supply current; RS grounded means normal mode. Add dominant bus current	Proper 60 Ω bus load at illustrative 2 V differential adds 33.3 mA while dominant; at 50% dominance, example total 26.7 mA. Reserve 72 mA from 17 + 3.3/60; this is a conservative circuit allowance, not a datasheet maximum
U801 AS5600	Current configuration is invalid; do not treat measured current as valid operation. Normal-mode datasheet ceiling 6.5 mA	Budget 6.5 mA after supply correction; low-power settings require separate timing assessment
U601 TSSP4038	About 0.35 mA typical, 0.45 mA maximum dark supply; illuminated/background behavior differs	Use 0.45 mA allowance for IC plus output pull-up load separately
U501 LM358	Exact BOM manufacturer's datasheet unverified. Legacy TI LM358 figures around 0.7 mA typical / 1.2 mA maximum are provisional family estimates, not a guarantee for C18164495	Remove for proposed current-sense redesign; INA180 supply 0.2 mA typical, 0.26 mA maximum, with small input-bias load from monitored rail

U306 BL8072	About 3 μ A typical / 5 μ A maximum ground current, plus all 3.3 V output load current	Same overhead if retained; heat, not quiescent current, is the concern
U301 LM1084	About 7 mA typical / 10 mA maximum fixed-version ground current, plus output load ; only relevant with UBEC logic path energized	Same if retained. Zero branch contribution assumed for disconnected UBEC in APM-only mode
D602/D605 TSAL6200, R603/R606 and D601/ D604	Each complete series branch has the same current, approximately $\max(0, (3.3 - V_{F\text{visible}} - V_{F\text{IR}}) / 1.2k)$; likely sub-mA, not a useful specified optical drive	Two 50 mA on-current emitters at 50% duty each give 50 mA combined average; up to 100 mA if both are on. Actual duty from validated firmware
Q602 TEFT4300 and R604	Light-dependent. With present 220 Ω pull-up, total supplied branch current ranges from the LED-loaded dark value (~3.18 mA example) toward 15 mA	With 10 k Ω , 0–0.33 mA. Datasheet photocurrent (0.8 mA minimum / 3.2 mA typical under specified irradiation) is not a fixed supply current
D603/R605	Included in analog branch above; approximately 3.18 mA dark example	Remove sensor-node load; optional separate indicator budget below

Q601, R601/ R602, D606/ R607	R601 collector path draws up to 3.3/4.7k $\approx$ 0.70 mA when low. Receiver high output cannot provide unrestricted base/LED current	Remove loaded inverter and LED branch. A 33 k Ω external pull-up, if used, adds up to 0.10 mA when output low
R501/R502	Current = $V_{shunt}/56.1k$: about 12.5 μ A at 7 A servo current	Remove with LM358 stage
R503/R504	Feedback leg current $\approx V_{out}/56.1k$; ideal 7 V operation is impossible here	Remove with LM358 stage
Q201/Q202, R202/R203/ R204	Transient/programming loads: each 10 k Ω base resistor can carry roughly (3.3–0.7)/10k $\approx$ 0.26 mA; EN pull-up up to 0.33 mA	Allocate 1 mA collectively; do not add transistor collector and its series resistor as separate loads
R201 4.7 k Ω	Up to 0.70 mA when COMMAND is driven to 3.3 V, generally supplied by the controller output	Include in system interface budget; not automatically from board regulator
R701/R702	Bus termination current already included in CAN calculation, not extra consumption per resistor	One fitted only at an endpoint; no double counting
Added I ² C pull-ups	Absent currently	Two 4.7 k Ω pull-ups draw at most 1.40 mA together when both low; use actual bus duty for average

Optional corrected visible indicators	Existing four LEDs counted in their branches above	Allow 2 mA each, up to 8 mA total, on independent drivers/resistors if all retained
D301/D302/D303	Carry the respective logic current; loss = $I \cdot V_F$, reverse leakage unknown until exact datasheet verified	Do not add carried current again; account for loss and junction heat
F1/F2, shunt 0.501, J1/J2/CN1, U302–U305	Carry load current; contact/fuse/shunt heat = $I^2 R$	No independent idle draw. Replacement resistance/current ratings must be included in voltage-drop and heat budget
C201–C206, C301–C304, C601/C602/C701/C801	Approximately zero ideal steady DC; actual leakage and dynamic charging depend on exact capacitor	Include effective capacitance, startup inrush and transient currents in regulator checks
J3, TP401/TP402, H101–H106	No intrinsic DC load	External programmer/load can change power flow; mechanical holes and test points do not consume current

APM and external UBEC	Output current is load-dependent; input quiescent current and efficiency curves not supplied	APM20 A is capability. Exact UBEC10 A is capability and is unsuitable for this input. Use measured conversion losses for battery budget
Tiankong H20/35 servo	Owner estimate: peak ~7 A at 7.4 V. Running/holding/stall duration and 5 V current unknown	Retain supplied peak for preliminary sizing; obtain exact manufacturer data and measure installed mechanism

Current references: [Espressif module](#), [ESP32 chip](#), [CH343](#), [CAN](#), [AS5600](#), [IR receiver](#), [phototransistor](#), [BL8072](#), [HTC LM1084](#), [provisional TI LM358 family reference](#).

Worked planning totals

These are explicitly **scenarios**, not measured consumption or guaranteed worst-case limits. The original circuit has faults, so its figure describes nominal powered electronics rather than successful operation.

Scenario, excluding servo	Original nominal circuit	Recommended corrected circuit
RF off, ESP32 assumed 50 mA; CAN 50% dominant at 2 V differential; encoder 6.5 mA; USB IC 3 mA	About 0.10 A on APM logic path	About 0.14 A with two emitters averaging 50 mA combined, corrected analog node and pull-ups

Same, but ESP32 TX average 239 mA	About 0.29 A	About 0.33 A
Conservative supply-capacity allowance including transients and uncertain small loads	0.65 A	0.75 A

Original worked example uses 40 mA for CAN with the unintended 40 Ω total bus load (one external 120 Ω assumed), analog 3.2 mA, LM358 0.7 mA, TSSP 0.35 mA and about 2 mA other loads. Corrected example uses 26.7 mA CAN, INA1800 0.2 mA, analog 0.33 mA, TSSP 0.45 mA, emitter average 50 mA and 3 mA miscellaneous/pull-up allowance. Optional indicators add up to 8 mA. UBEC operation through LM1084 adds approximately 7–10 mA to input current, plus its voltage-dependent power loss. These examples deliberately exclude unidentified external 2–3 A loads.

For preliminary sizing at the supplied 7 A servo peak, the existing linear architecture would need approximately **7.65 A**, and the corrected design budget **7.75 A**, at the low-voltage converter output. This assumes the logic branch is fed from that same source. The original circuit is not capable of using that current correctly; its shunt would dissipate 4.9 W. The corrected estimate is below 10 A but does not make the specified UBEC compatible with 78 V or 15 A continuous.

At 5 V, use **I_{servo}(5 V) + up to 0.75 A**; do not assume the 7.4 V peak also describes 5 V. If 15 A means the total converter output, reserve 0.75 A for electronics and 14.25 A for servo/external load. If 15 A must remain available to the servo alone, size the source for at least 15.75 A plus the relevant margin.

Battery current must be calculated by power. For a hypothetical suitable 7.4 V converter retaining the linear logic branch, 7.75 A output at 78 V and assumed 90% efficiency gives $7.4 \times 7.75 / (78 \times 0.90) \approx \mathbf{0.82\ A}$ **input**, plus any converter idle current omitted by that efficiency assumption. At 15 A servo plus 0.75 A logic, the same calculation gives about **1.66 A**. The 5 V APM at 15.75 A output gives approximately **1.12 A** under the same illustrative efficiency. Battery current rises as battery voltage falls. These are not APM efficiency guarantees, and the

unsuitable10A UBEC is not endorsed by the calculation.

Automated check results and release conditions

Original ERC: **81 findings**, dominated by custom-library/pin-type issues, plus the missing LM358 B unit. Original PCB DRC: **75 findings**, including **10 starved-thermal errors**, **0 unconnected items**, and **46 separate schematic-parity findings concerning datasheet fields**.

The 46 metadata mismatches are not 46 proven connectivity faults.

Library/footprint mismatches require review because some custom numbering is intentional; do not blindly overwrite footprints from standard libraries. Six padstack warnings concern paste reduction on F1/F2 and the shunt; resolve for intended stencil or document hand assembly.

Silkscreen/text warnings are lower priority than electrical and thermal errors. Some checks were disabled in the project; zero unconnected items is not a complete manufacturability pass. See the saved JSON reports for exact locations.

Before a functional release:

1. Implement the circuit corrections in a new electrical revision; keep schematic, PCB and both assembly variants consistent. Regenerate manufacturing outputs only after this work. The annotated review copy supplied here has **not** implemented them.
2. Obtain the missing exact component and mechanical information, complete pin/package and insulation checks, and resolve all substantive ERC/DRC findings. Preserve the casing outline unless an agreed mechanical change is made.
3. Bring up logic on a current-limited low-voltage supply first. Verify every rail, programming, encoder communication/magnet status, CAN termination and control-signal levels before attaching the servo or raw battery.
4. Measure servo running, starting, holding and brief jam currents at both permitted voltages. Verify current-sense scaling against an independent instrument; demonstrate prompt, safe jam handling without brownout or unintended movement.

5. Load-test each intended 15 A path to thermal equilibrium inside the actual enclosure, at worst ambient. Measure copper, contacts, fuses, shunt, standoffs, regulators and wiring. Require copper temperature rise $\leq 15\text{--}20^\circ\text{C}$ and every component within its own derated limits. Measure rail sag/overshoot during servo steps. A copper temperature target does not mean regulator junctions may rise only 20°C .
6. Test source selection and all powered/unpowered/programmer combinations for backfeed. Check raw-input startup/hot-plug and transient protection with appropriate test equipment and limits.
7. Register the DXF/casing/gear assembly to the PCB; verify component heights, magnet gap, sensor axes and swept gear clearance. Run both bait sizes across the full opening, with sunlight/dust/vibration and worst speed, logging missed/double counts and sensor faults. Confirm disarmed and lost-command behavior in firmware.

The highest-priority functional path is to qualify the 5 V APM build, correct current sensing, AS5600, IR and CAN circuits, and upgrade the complete high-current path. The 7.4 V option additionally requires a replacement converter suitable for 18S/78 V and the specified continuous load.